

In 1938 C.E. Shannon in his paper "A Symbolic Analysis of Relay and Switching Circuits" introduced a two-valued Boolean algebra called Switching algebra, in which he demonstrated that the properties of bistable electrical switching circuits can be represented by this algebra.

For the formal definition of Boolean algebra, we shall employ the postulates formulated by E.V. Huntington in his 1904 paper "Sets of Independent Postulates for the Algebra of Logic."

These postulates or axioms are not unique for defining Boolean algebra. For reference read "Modern Applied Algebra" by Birkhoff and Boute.

Boolean algebra is an algebraic structure defined on a set of elements B together with two binary operators $+$ and $\cdot$ provided the following (Huntington) postulates are satisfied:

1. (a) Closure with respect to the operator $+$.
2. (a) Closure with respect to the operator $\cdot$.
2. (a) An identity element with respect to $+$, designated by 0 : $x+0=0+x=x$.
2. (b) An identity element with respect to $\cdot$, designated by 1 : $x \cdot 1 = 1 \cdot x = x$.
3. (a) Commutative with respect to $+$: $x+y=y+x$.
3. (b) Commutative with respect to $\cdot$: $x \cdot y = y \cdot x$.
4. (a) $\cdot$ is distributive over $+$: $x \cdot (y+z) = (x \cdot y) + (x \cdot z)$.
4. (b) $+$ is distributive over $\cdot$: $x + (y \cdot z) = (x+y) \cdot (x+z)$.
5. For every element $x \in B$, there exists an element $x' \in B$ (called the complement of x) such that: (a) $x+x'=1$ and (b) $x \cdot x'=0$.
6. There exists at least two elements $x, y \in B$ such that $x \neq y$.

In order to have a Boolean Algebra, one must show that:

1. the elements of the set B .
2. the rules of operation for the two binary operators and
3. that the set of elements B together with the two operators, satisfies the six Huntington postulates.